

DATA STRUCTURE ALGORITHM AND OBJECT ORIENTED PROGRAMMING IN C++

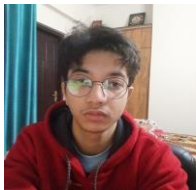
DSCPP-III-2026-T162

PROJECT AND TEAM INFORMATION

Project Title

Online Train Ticket Booking System

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>Vibe coders</i>
Team member 1 (Team Lead) <i>(Last Name, name: student ID: email, picture):</i>	<i>Chandola</i> <i>Omansh –</i> <i>7895739833</i> <i>2510012482@geu.ac.in</i> 
Team member 2 <i>(Last Name, name: student ID: email, picture):</i>	<i>Singh Vineet –</i> <i>8840423711</i> <i>Vineetsingh55555rbl@g</i> <i>mail.com</i> 

Team member 3 (Last Name, name: student ID: email, picture):	<i>Bhandari raj Yash -</i> <i>75790 54881</i> <i>yashrajbhandari56</i> <i>1@gmail.com</i> 
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PROPOSAL DESCRIPTION

Motivation

The motivation of behind the developments of the online train ticket booking system is to understand real world railroad reservation , ticket cancelation and waiting list with the implementation of C++ and DSA(data structure and algorithms) , this project of C++ and DSA topics of file handling , classes and objects , sorting , queue ,LinkedList , this project also help in understanding logical thinking , problem solving , and to design a computer application software , overall this project is help in connecting the practical knowledge with real life application creating a ticket booking software more efficient and organized .

This also helps in developing the coding concepts of C++ and DSA (Data Structures and Algorithms) helps in improving the errors.

State of the Art / Current solution

Currently, online train ticket booking in India is mainly handled through platforms such as **IRCTC** and **IXIGO**, which provide services for train search, route and fare information, seat availability, ticket booking, and cancellation. We modernized the system by making booking easier for the user user can check the waiting list, available seats, best train available at best price for the passenger and cancel reservations view their booking details

While designing the system we use concepts of DSA in c and fundamentals of C++ concepts such as stack, queue, LinkedList etc, and in C++ we use concepts such as class, object, inheritance, polymorphism and concepts of array, dynamic memory allocation to manage train and passenger information respectively to organize and store important booking information.

Trough this project we learned the combine C++ and DSA programming skills to develop a practical application. It improves the problem solving, team work, research abilities and understanding on how data manage in real-life system.

Project Goals and Milestones

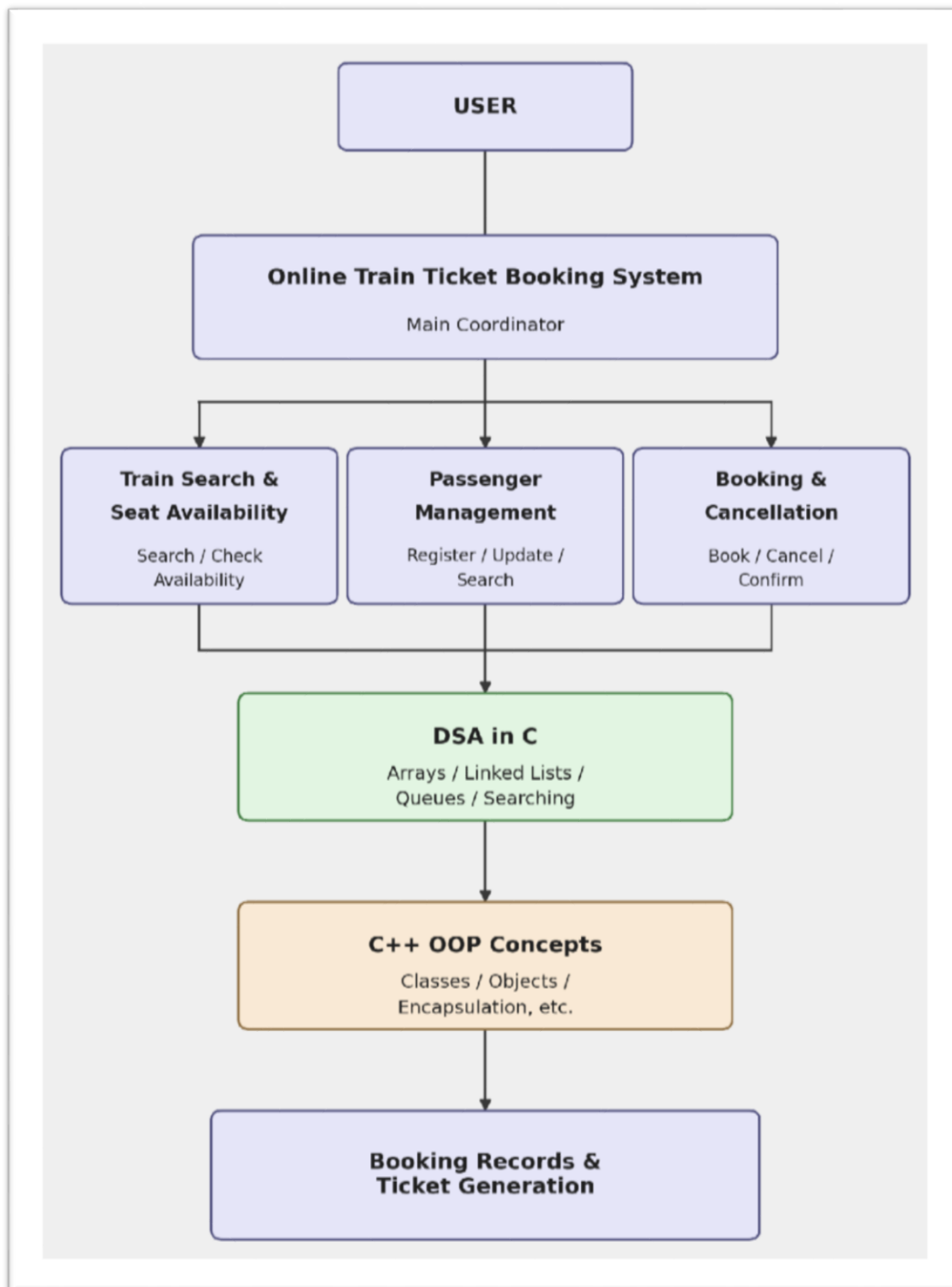
Goals: The main goal of our project of online train ticket booking is help us to develop the understanding the concepts of C++ and data structure algorithms and develop a software for develop a simple efficient and user-friendly railway reservation platform. The system is made for train searching, train booking, easier and less dependency on other platforms for train booking.

Milestone: system designing of overall structure, modules, menu and data applications use of array, stack, queue, LinkedList use of class, objects, inheritance and file handling develop the functionality train and check availability seats add ticket cancellation and seat update functionality, combinations of different modules in one system, prepare and demonstration of projects

Project Approach

- The online railway train ticket booking system will be developing using C++ and data structures and algorithm (DSA) the project help to make a menu driven application
- Use of suitable DSA concepts such as arrays, structure, queues, stacks, searching and sorting.
- Develop the individual modules for train management, passenger management, booking and cancellation.
- Use of class, objects and inheritance and file handling to develop a system.
- Check the system correctness and proper implementation of C++ and DSA
- And check final working model.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The main outcome of the project is to develop a train ticket booking system using C++ and Data structures and algorithms the system will provide an organised a platform to search train, book a seat, views on waiting list, availability, schedule of train, cancel reservation and booking details.

This concepts help in the understanding the basics of C++ and DSA (Data structure and algorithms this we will make the ticket booking convenient and easier this will help in deep understanding of the knowledge of the topic.

DSA is used to improve the data management and system efficieny for maintain the large amount of data from the user for booking the train ticket. So that system have well known about the amount of ticket booked and total number of passengers travelling in a train and total number of seats available in a train and concepts of object oriented programming leads to desining a class and objects .

Assumptions:

The assumptions of this projects while developing online ticket booking system:

1. This project help in understanding concepts of DSA and C++.
2. User should provide a valid booking id for the ticket.
3. Every person has a user booking ID.
4. In this train is a class and passenger is a object.
5. Train details such as train number, timing, seat capacity in a predefined system.
6. The system should have menu driven applications.
7. Data should be stored properly.
8. The system should have a real time railway reservation.

Reference:

- Object oriented programming in C++ using topics like class , object , inheritance, instance, polymorphism and encapsulation.
- IRCTC – government website for booking train ticket
- W3schools – for basics and fundamentals of c++
- Tutorial points – concepts of object oriented programming
- Geeks for geeks – for the knowledge of array , stack , queue , linkedlist and advance concepts of DSA
- Fundamentals of c – by Dennis Ritchie.